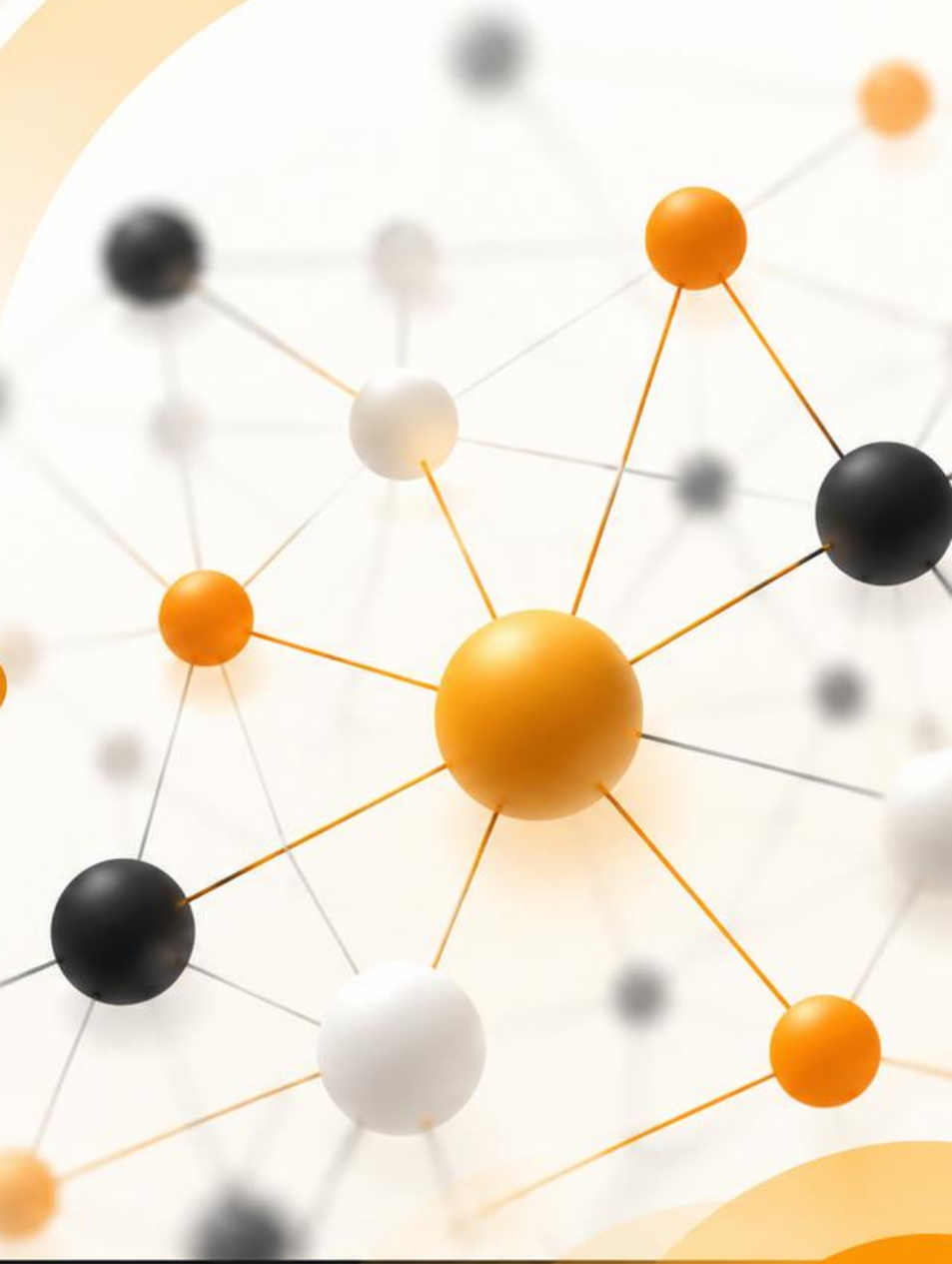
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Why Do We Need Property Graphs?

Understanding when connected data needs a graph model

Bilal Ben Mahria



CONNECTED DATA. BRIGHTER ANSWERS.

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The Problem:
When Relationships Become First-Class Citizens

1 A Familiar E-commerce Database

You are a data engineer at a mid-sized e-commerce company.
Your PostgreSQL database has been serving the business well for years, storing customers, orders, products, and categories.



Customers, orders, products, categories — all normalized, indexed, query-optimized.

2 The Business Question

Then the product team asks you to build:



> "Show me all products that were bought by **customers who also bought products that are frequently co-purchased** with the products Alice has bought."

3 The SQL Reality

You open your SQL client. After 20 minutes, you have a query with seven JOINs, two CTEs, and a DISTINCT clause you are not fully confident about.

```sql
WITH alice_products AS ( ... ),
co_purchased AS ( ... ),
other_customers AS ( ... )

SELECT DISTINCT p.*
FROM customers c
 JOIN orders o1         ON c.customer_id = o1.customer_id
 JOIN order_items oi1   ON o1.order_id = oi1.order_id
 JOIN products p1       ON oi1.product_id = p1.product_id
 .. more JOINs (total 7) ...
```

20 minutes
to write the query

7 JOINs
+ 2 CTEs
+ DISTINCT

4 million rows
touched for Alice alone
(nested loop join)

And now imagine running this for **100,000 active customers** simultaneously to power a homepage recommendation widget.

4 Why This Happens

Relational Model (Row-based)



- Data is stored in rows and tables
- Relationships are represented by foreign keys (indirect pointers)
- Traversing multiple relationships requires expensive JOIN operations

Property Graph Model



- Data is stored as nodes and relationships
- Relationships are first-class, directly navigable structures
- The deeper the traversal, the larger the performance and modeling advantage

5 This is a Real-World Challenge

This is not a made-up scenario. It is the daily reality of teams trying to build **recommendation engines**, **fraud detection systems**, **supply chain risk analyzers**, and **social network features** on top of relational databases that were never designed for this class of question.

6 Key Takeaway

The deeper the relationship traversal, the larger the performance and modeling advantage of the graph.

7 What We Will Learn

What is a **property graph**, why was it invented, and when does it genuinely earn its place in your architecture?

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BRIGHTER TOMORROW THROUGH DATA

The Problem: When Relationships Become First-Class Citizens

TURNING DISCONNECTED DATA INTO MEANINGFUL CONNECTIONS

A Real-World Scenario

You are a data engineer at a mid-sized **e-commerce company**. Your PostgreSQL database has served well for years. It stores customers, orders, products, and categories — all normalized, indexed, and query-optimized.

> Show me all **products** that were bought by **customers** who also bought products that are frequently co-purchased with the products Alice has bought."



In a Relational Database (PostgreSQL)

After 20 minutes, you have a query with seven JOINs, two CTEs, and a DISTINCT clause you are not fully confident about.

```sql
WITH alice_products AS (
  SELECT p.id AS product_id
  FROM orderso
  JOIN order_items oi ON oi.order_id = o.id
  JOIN products p ON p.id = oi.product_id
  JOIN customers c ON c.id = o.customer_id
  WHERE c.name = 'Alice'
), co_purchased AS (
  SELECT p2.id AS product_id, COUNT(*) AS cnt
  FROM order_items oi1
  JOIN order_items oi2 ON oi1.order_id = oi2.order_id
  JOIN products p2 ON p2.id = oi2.product_id
  WHERE oi1.product_id IN (SELECT product_id FROM alice_products)
    AND oi1.product_id <> oi2.product_id
  GROUP BY p2.id
)
SELECT DISTINCT p.*
FROM co_purchased cp
JOIN products p ON p.id = cp.product_id
ORDER BY cp.cnt DESC
LIMIT 20;
```

7 JOINs
2 CTEs
DISTINCT

Nested loop join

4 million rows for Alice alone

100,000 active customers?

In a Property Graph Database

Relationships are stored as first-class citizens. The same question becomes a natural graph traversal.

```
MATCH (alice:Customer {name: 'Alice'})-[:BOUGHT]->
      (p:Product)-[:CO_PURCHASED_WITH]->(rec:Product)
     <-[:BOUGHT]-(other:Customer)
WHERE other <> alice
RETURN DISTINCT rec
ORDER BY rec.popularity DESC
LIMIT 20
```



This Is Not a Toy Example

This is the daily reality of teams trying to build:

Recommendation engines
"Customers who bought X also bought Y"

Fraud detection systems
Detecting suspicious patterns across linked entities

Supply chain risk analyers
Tracing dependencies across suppliers, products, and regions

Social network features
Finding people, content, or communities through connections

The Root Cause Is Architectural

Relational databases store data in rows and relate it through foreign keys — **indirect pointers** to other rows. This is excellent for many workloads but becomes expensive at depth.

→

Property graph databases store relationships as first-class, **directly navigable structures.** The deeper the traversal, the larger the performance and modeling advantage.

What This Module Covers

It answers the foundational question:

> What is a property graph, why was it invented, and when does it genuinely earn its place **in your architecture?**

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KNOWLEDGE GRAPHS FOR A BRIGHTER TOMORROW

Worked Example: Alice's Neighborhood — SQL Version (Relational Database)

TURNING DISCONNECTED DATA INTO MEANINGFUL CONNECTIONS

Business Question

"Which products were purchased by customers who bought products similar to the products purchased by Alice?"

4-Hop Logic

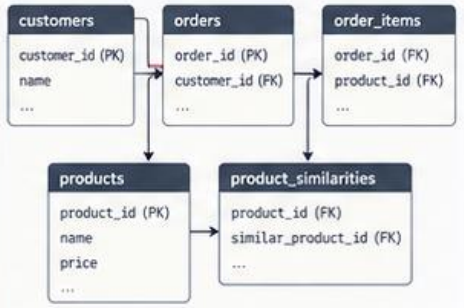
1. Alice → her Orders
2. Orders → Products she bought
3. Those Products → Similar Products
4. Similar Products ← Orders ← Other Customers

SQL Solution (using CTEs)

PostgreSQL / Relational Database

```
-- Step 1: Find products Alice purchased
WITH alice_products AS (
    SELECT DISTINCT oi.product_id
    FROM customers c
    JOIN orders o ON c.customer_id = o.customer_id
    JOIN order_items oi ON o.order_id = oi.order_id
    WHERE c.name = 'Alice Chen'
),
-- Step 2: Find products similar to what Alice bought
similar_products AS (
    SELECT DISTINCT ps.similar_product_id
    FROM product_similarities ps
    WHERE ps.product_id IN (SELECT product_id FROM alice_products)
),
-- Step 3: Find customers who bought those similar products
customers_who_bought_similar AS (
    SELECT DISTINCT o.customer_id
    FROM orders o
    JOIN order_items oi ON o.order_id = oi.order_id
    WHERE ps.product_id IN (SELECT similar_product_id FROM similar_products)
    AND o.customer_id != (SELECT customer_id FROM customers
        WHERE name = 'Alice Chen')
),
-- Step 4: Find all products those customers bought
recommended_products AS (
    SELECT DISTINCT p.product_id, p.name, p.price,
           COUNT(DISTINCT o.customer_id) AS buyer_count
    FROM customers c
    JOIN orders o ON c.customer_id = o.customer_id
    JOIN order_items oi ON o.order_id = oi.order_id
    JOIN products p ON oi.product_id = p.product_id
    WHERE c.customer_id IN (SELECT customer_id FROM customers_who_bought_similar)
    AND p.product_id NOT IN (SELECT product_id FROM alice_products)
    GROUP BY p.product_id, p.name, p.price
)
SELECT product_id, name, price, buyer_count
FROM recommended_products
ORDER BY buyer_count DESC
LIMIT 20;
```

Relational Schema (Simplified)



What is a CTE?

A CTE (**Common Table Expression**) is a temporary named result set that you can reference within a query. It helps break complex queries into readable steps.

Observations about the SQL version

1. 4 CTEs required to keep it readable (without CTEs, this is a deeply nested subquery)
2. 8 total table references (customers appears twice, orders twice, order_items twice)
3. The query engine must materialize intermediate result sets
4. Adding another hop (e.g., "products also rated highly by those customers") requires another CTE and JOIN
5. Query complexity scales with hop depth, not just data volume

Key Takeaway

In SQL, connecting data requires **multiple JOINs** and **intermediate result sets.** As the number of hops increases, the query becomes more complex, harder to read, and more expensive to execute.

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Worked Example: Alice's Neighborhood — Cypher Version (Property Graph)

TURNING DISCONNECTED DATA INTO MEANINGFUL CONNECTIONS

Business Question

"Which products were purchased by customers who bought products similar to the products purchased by Alice?"

4-Hop Logic

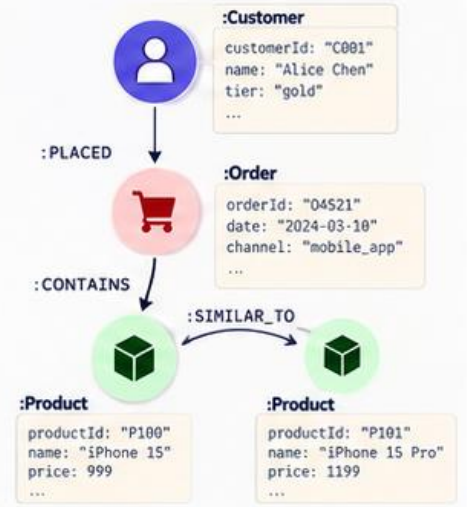
1. Alice → her Orders
2. Orders → Products she bought
3. Those Products → Similar Products
4. Similar Products ← Orders ← Other Customers

Cypher Solution (Property Graph)

Neo4j / Property Graph Database

```
// Find products bought by customers who bought products similar to Alice's products
MATCH (alice:Customer {name: 'Alice Chen'})
      -[:PLACED]->(:Order)
      -[:CONTAINS]->(aliceProduct:Product)
      -[:SIMILAR_TO]->(similarProduct:Product)
      <-[:CONTAINS]-(:Order)
      <-[:PLACED]-(otherCustomer:Customer)
      -[:PLACED]->(:Order)
      -[:CONTAINS]->(recommended:Product)
WHERE otherCustomer <> alice
  AND NOT (alice)-[:PLACED]->(:Order)-[:CONTAINS]->(recommended)
RETURN recommended.name AS productName,
       recommended.price AS price,
       COUNT(DISTINCT otherCustomer) AS buyerCount
ORDER BY buyerCount DESC
LIMIT 20
```

Graph Model (Simplified)



Visual Traversal (4 Hops)



Observations about the Cypher version

1. The query reads like a description of the traversal path
2. No explicit JOIN syntax — relationships are navigated directly in the MATCH clause
3. Adding another hop is structural, not syntactically explosive
4. The WHERE clause handles exclusions (not already purchased, not Alice herself)
5. The query expresses the shape of the data we are looking for

Key Takeaway

In a property graph, relationships are first-class and directly navigable. The query is more intuitive, easier to extend, and scales naturally with the depth of the traversal.

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OLTP vs OLAP

A beginner-friendly comparison

Same data. Different purposes. Both are important.

OLTP | Online Transaction Processing

Built for **day-to-day operations** and **many small, fast transactions**.



Main goal	**Run day-to-day business operations** reliably and accurately.
Typical users	**Frontline staff, customers, operational teams** (e.g., cashiers, tellers, customer service).
Type of queries	**Simple, short, predefined queries** (e.g., insert, update, delete, select one record).
Data focus	**Current, up-to-date** operational data at the **row/record level**.
Speed expectation	**Very fast** (milliseconds).
Example systems	E-commerce systems, banking systems, POS (point of sale), airline/booking systems.
Example question	"Place an order", "Withdraw money from an ATM", "Update inventory", "Book a ticket".

OLAP | Online Analytical Processing

Built for **analysis, reporting, dashboards,** and **finding trends** in large amounts of data.



Main goal	**Analyze data**, create reports and dashboards, and **find trends** to support better decisions.
Typical users	**Analysts**, business users, data scientists, management.
Type of queries	**Complex, long-running analytical queries** with aggregations (e.g., SUM, AVG, GROUP BY).
Data focus	**Historical and summarized data** (often from months or years of data).
Speed expectation	**Slower** (seconds to minutes), but handles large amounts of data.
Example systems	Data warehouses (e.g., Snowflake, BigQuery, Redshift), BI tools (e.g., Power BI, Tableau).
Example question	"What were monthly sales by region?", "What are the top-selling products this quarter?", "What are customer trends over 2 years?", "Show me a revenue dashboard".

Common OLTP Examples



Placing an order



Withdrawing money from an ATM



Updating inventory



Booking a ticket

Aspect	OLTP	OLAP
Read/Write pattern	Many small writes and reads	Fewer but larger queries
Data timeframe	Current operational data	Historical and summarized data
Level of detail	Row-level (individual records)	Aggregations (SUM, AVG, GROUP BY)
Typical data size	Transaction-by-transaction	Large volumes of data

Key Terms (in simple language)

Transaction	A small unit of work such as placing one order or updating one record.
Analytics	Studying data to find patterns, summaries, and insights.
Dashboard	A screen with charts and KPIs that help people monitor the business.
KPI	Key Performance Indicator — a metric used to track performance, such as revenue or conversion rate.

OLTP runs the business. OLAP helps you understand the business.

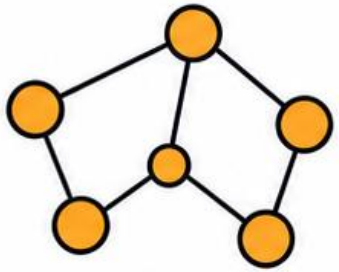
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C. Conceptual Explanation

From Graph Theory to the Property Graph Model

1 C.1 What Is a Graph?

A graph is a structure made of **vertices (nodes)** and **edges (links)** connecting pairs of vertices.



- **Vertices** = nodes or points
- **Edges** = links or arcs between vertices

Graphs are one of the most powerful and general data structures in computer science.

2 Where Graphs Appear



Family trees



Road networks



The internet



Social networks



Electrical circuits



Dependency graphs

Many real-world systems are naturally modeled as graphs.

3 Common Graph Variants

Undirected

Edges have no direction.



Example: friendships

Directed

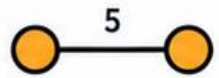
Edges have direction.



Example: A follows B

Weighted

Edges carry a value.



Example: distance or strength

Labeled

Nodes and/or edges carry type information.



Example: relationships with types

4

From abstract graph theory to a practical database model

5 C.2 The Property Graph Model

A **property graph** is a pragmatic specialization of the labeled directed graph, designed for database systems.

A Nodes

- Represent entities in the domain
- Have one or more **labels**
- Carry **properties** (key-value pairs)

Node example

```
Labels: [:Customer]
Properties: {
  customerId: "C001",
  name: "Alice Chen",
  email: "alice@example.com",
  tier: "gold",
  registeredAt:
  datetime("2023-01-
  15T10:30:00")
}
```

B Relationships

- Represent connections between nodes
- Are always **directed**
- Carry exactly one type
- Also carry properties

Relationship example

```
Type: :PURCHASED
Start: Customer (Alice Chen)
End: Order (#4521)
Properties: {
  purchasedAt:
  datetime("2024-03-10T14:22:00"),
  channel: "mobile_app"
}
```

C Labels

- Labels are type tags on nodes
- A **node** can have multiple labels

Node with multiple labels

```
(:Customer:VIPMember
 {customerId: "C001",
  vipSince: 2022})
```

Key Insight

Relationships in a property graph are not just foreign key pointers.
They are first-class data structures with their own identity, type, direction, and properties.

This is the fundamental architectural **difference** from relational databases.

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The Relational Model

Data in tables, related by keys, queried with SQL — reliable, mature and powerful.

TURNING DISCONNECTED DATA INTO MEANINGFUL CONNECTIONS

Core Concepts

1. **Data in Relations (Tables)**
   Data is stored in tables (rows and columns). Each row represents one fact about an entity or a relationship.

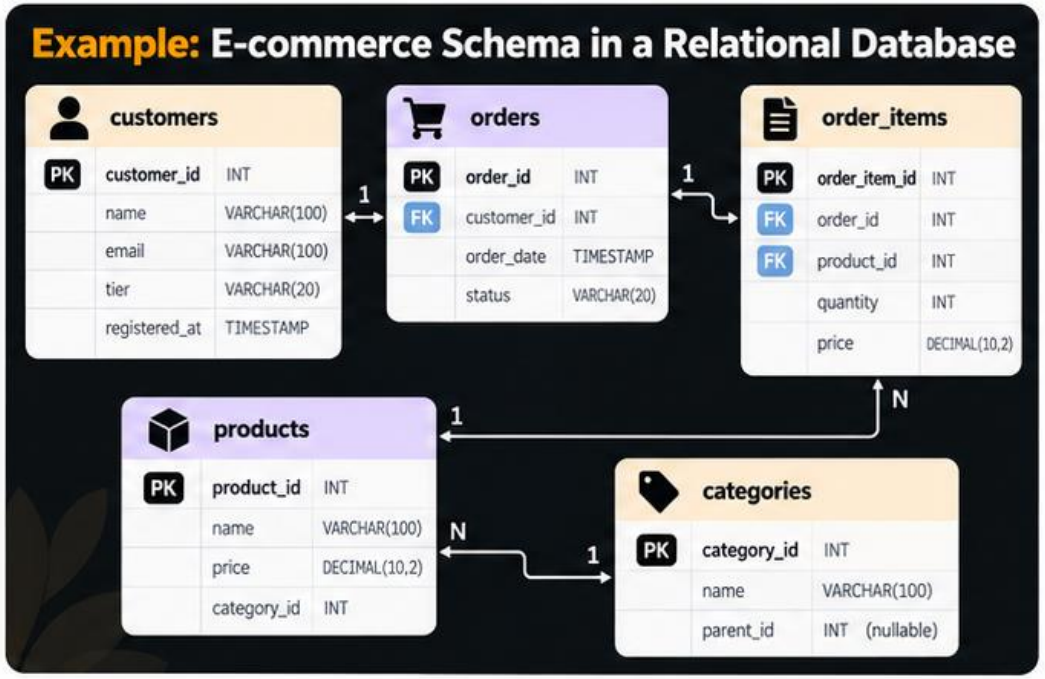
2. **Foreign Keys**
   Relationships are represented by foreign keys. A `product_id` in `order_items` points to the products table. The link is resolved at query time.

3. **Schema is Enforced**
   Every row must conform to the table's definition (data types, nullability, unique constraints, etc.).

4. **Normalization**
   Data is decomposed into multiple tables to eliminate redundancy and keep single-fact responsibilities.

5. **SQL Query Language**
   SQL is declarative: you describe what you want, not how to compute it. It is optimized for set-based operations over entire tables or subsets.

Example: E-commerce Schema in a Relational Database



How Foreign Keys Work

`order_items.product_id` is a foreign key that references `products.product_id`.

order_items

order_item_id	product_id	quantity
1001	42	2

products

product_id	name	price
42	Wireless Headphones	199.00

Sample SQL Query

Find all products purchased by Alice Chen:

```
SELECT p.product_id, p.name, p.price
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products p ON oi.product_id = p.product_id
WHERE c.name = 'Alice Cheh';
```

ACID Transactions: Strong Consistency Guarantees

A transaction is a group of operations executed as a single unit of work. ACID guarantees ensure data remains consistent, even in case of errors or concurrent access.

A — Atomicity
All operations succeed or all fail.
Example: Transfer money from account A to B. Either both debit and credit happen, or neither does.

C — Consistency
Moves the database from one valid state to another.
Example: A foreign key constraint ensures you cannot insert an order with a non-existent customer_id.

I — Isolation
Concurrent transactions do not interfere with each other.
Example: Two users updating the same product stock see a consistent result.

D — Durability
Once committed, changes persist even if the system crashes.
Example: After an order is confirmed, it remains in the database, even after a power failure.

Strengths

- Mature tooling, deep ecosystem, widely understood
- Strong consistency guarantees (ACID transactions)
- Excellent for aggregate queries over large datasets
- Enforced schema prevents a class of data quality problems
- Optimal for tabular (OLTP) and column-optimized (OLAP) workloads

Weaknesses

- JOIN cost grows with table size; deep multi-table joins are expensive
- Schema rigidity makes iterative modeling expensive
- Recursive or variable-depth relationships require workarounds (adjacency list, path enumeration, closure table, recursive CTEs)
- Graph-shaped questions produce exponentially complex SQL

Key Takeaway

The **relational model** is the foundation of modern data systems: reliable, consistent, and excellent for tabular data.
However, as the depth and complexity of relationships grow, SQL queries become harder to write, understand, and scale.

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The Labeled Property Graph Model

Data as a network of **nodes** and relationships — flexible, expressive, built for connected data.

TURNING DISCONNECTED DATA INTO MEANINGFUL CONNECTIONS

Core Concepts

1 Data is a Network

Data is modeled as a network of nodes connected by relationships.

2 Nodes

Represent entities, have one or more labels, and carry properties (key-value pairs).

3 Relationships

Represent connections between nodes. They are typed, directed, and carry properties — first-class objects in the data model.

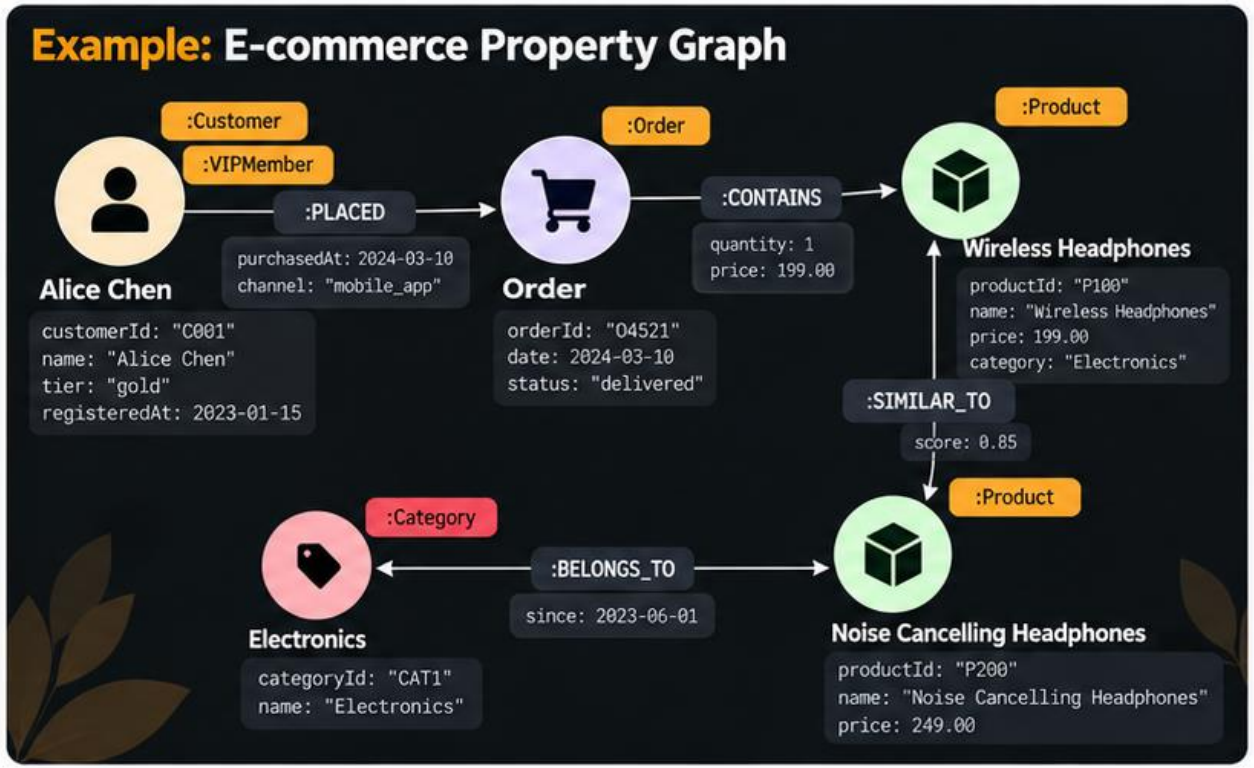
4 Schema is Optional

Nodes can have any combination of labels and properties. No rigid column definitions.

5 Cypher (and GQL)

Query language based on pattern matching over graph structure. Cypher is widely used today, and ISO GQL is emerging as the standard.

Example: E-commerce Property Graph



Node vs. Relationship Example

A Node

:Customer
:VIPMember

customerId	"C001"
name	"Alice Chen"
email	"alice@example.com"
tier	"gold"
registeredAt	2023-01-15T10:30:00

A Relationship

:PURCHASED

From	(:Customer)
To	(:Order)
Type	:PURCHASED
Properties	
purchasedAt	2024-03-10T14:22:00
channel	"mobile_app"

Cypher Query Example

Find products purchased by customers who bought products similar to Alice's products

```
MATCH (alice:Customer {name: 'Alice Chen'})
      -[:PLACED]->(:Order)
      -[:CONTAINS]->(aliceProduct:Product)
      -[:SIMILAR_TO]->(similarProduct:Product)
      <-[:CONTAINS]-(:Order)
      <-[:PLACED]-(otherCustomer:Customer)
      -[:PLACED]->(:Order)
      -[:CONTAINS]->(recommended:Product)
WHERE  otherCustomer <> alice
      AND NOT (alice)-[:PLACED]->(:Order)-[:CONTAINS]->(recommended)
RETURN recommended.name AS productName,
      recommended.price AS price,
      COUNT(DISTINCT otherCustomer) AS buyerCount
ORDER BY buyerCount DESC
LIMIT 20
```

Strengths

- Natural fit for relationship-intensive, traversal-heavy workloads
- Variable-depth path queries are first-class features (not hacks)
- Schema flexibility enables rapid iteration
- Relationship properties eliminate junction-table complexity
- Intuitive for domain modeling (whiteboard diagrams map directly to graph schema)

Key Terms (Quick Definitions)

Node	An entity in the graph (e.g., a customer, product, order).
Label	A tag that classifies a node (a node can have multiple labels).
Property	A key-value pair on a node or relationship.
Relationship	A directed, typed connection between two nodes.
Type	The name of a relationship (e.g., :PURCHASED, :SIMILAR_TO).
Cypher	Graph query language used by Neo4j.
GQL	ISO standard for graph query language (Cypher-compatible).

Weaknesses

- Weaker out-of-the-box schema enforcement (APOC/constraints help, but it's not relational-level)
- Bulk aggregations over entire graphs are slower than column-store databases
- Less mature ecosystem than PostgreSQL (fewer BI tools, less SQL interoperability)
- Not a good choice for pure OLAP/reporting workloads
- Horizontal scaling is more complex than relational databases at large scale

Key Takeaway

The labeled property graph model makes relationships first-class citizens, enabling natural and efficient traversal of connected data. It is highly expressive and flexible, but it is not a replacement for analytical databases.

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Why Is Traversal Fast in a Graph Database?

JOINs vs. Index-Free Adjacency

In a Relational Database

Connecting data requires a **JOIN** — a computational operation that scans index structures and matches rows from different tables.

The **cost of joins grows with data volume**.

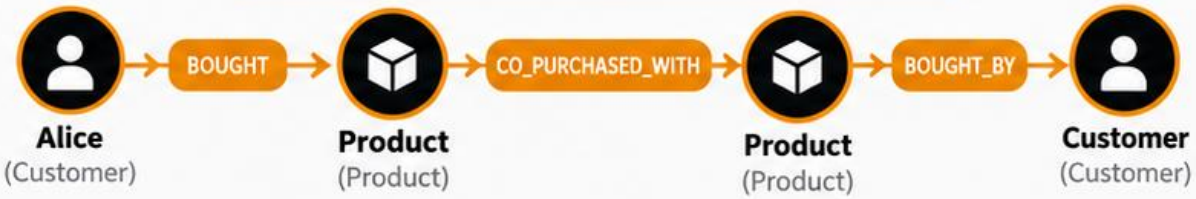
Customer		JOIN	Order		JOIN	OrderItem		JOIN	Product	
customer_id	PK	⟷	order_id	PK	⟷	order_item_id	PK	⟷	product_id	PK
name			customer_id	FK		order_id	FK		name	
email			order_date			product_id	FK		category	
...			...			quantity			price	
						...			...	

Search + match rows across tables
Cost grows with scale and join depth.

In a Property Graph Database

Each node stores direct pointers (internal record IDs) to adjacent relationships. Following a relationship is a **constant-time pointer dereference**, not a search operation.

Index-Free Adjacency



Direct traversal from one connected record to the next.

- Local traversal
- Pointer dereference
- Efficient deep traversal on connected data

Core Insight

Index-free adjacency is the core reason deep traversals can outperform equivalent multi-join SQL on connected data.

Honest Caveat

Index-free adjacency mainly benefits **local traversals** from a starting node. For **global aggregations** — such as counting all products by category across the entire database — relational databases, especially optimized column stores, can perform comparably or better.

Choose graphs for deep connected traversals; do not assume graphs are automatically best for every global analytical query.

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Labels in the Property Graph Model

Labels are **tags** that describe the role of a node — they are **not** a type hierarchy.

TURNING
DISCONNECTED DATA
INTO MEANINGFUL
CONNECTIONS

1 Labels as Tags

Labels act as **tags** that describe the role of a node in the graph. A node can have one or more labels.

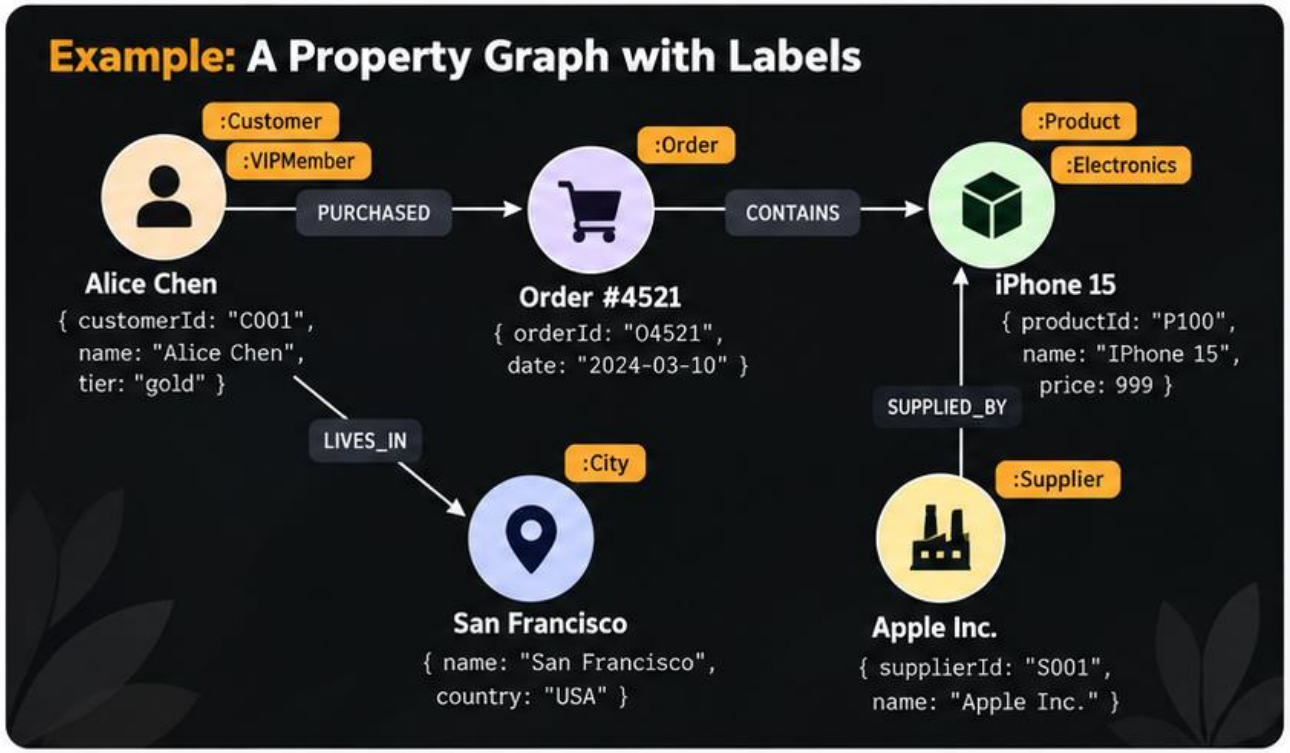
```
(:Customer)
(:Product:Electronics)
(:User:VIPMember)
```

2 No Associativity

Labels do not have inherent relationships. You cannot infer hierarchies between labels.

✖ `(:Apple)` is not inherently a specialization of `(:Fruit)`

Example: A Property Graph with Labels



3 Feature, Not a Bug

The lack of associativity is intentional. Type systems and polymorphism are handled at the **application level**, not the database level.

✔ **Labels are simple tags, by design.** This keeps the graph model flexible and practical.

4 Separation of Concerns

The labeled property graph model separates data and type systems.

Database
Owns the data and its structure (nodes, relationships, labels, properties).

Application
Can define its own type system, business logic, and hierarchies for specific needs.

5 Labels vs. Types

Labels do not function as types with inheritance. They are simply indicators of a node's role(s) in the graph.

✔ What Labels ARE

- Tags describing a node's role
- Multiple labels per node
- No inherent meaning beyond being a tag
- Used for filtering and grouping in queries
- Simple and flexible

✖ What Labels are NOT

- Not a type hierarchy
- Not connected to each other (no "is-a" relationships)
- Not classes with inheritance
- Not a mechanism for polymorphism
- Not enforcing business logic

6 Application-Level Decision

If you map labels to types (e.g., in your application code), it is a design choice made by the system implementer, not something provided or enforced by the graph database.

Key Takeaway

Labels are lightweight, non-hierarchical tags that classify nodes.
They provide flexibility and keep the graph model simple, while **type** systems and hierarchies are handled at the application level.

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The RDF Model

A universal, interoperable model for the Semantic Web — everything is a triple.

TURNING DISCONNECTED DATA INTO MEANINGFUL CONNECTIONS

Core Concepts

1. **Everything is a Triple**
   Data is represented as a set of statements in the form:
   Subject → Predicate → Object
   Each triple expresses one fact.

2. **URIs Identify Everything**
   Subjects, predicates and (most) objects are identified by URIs (Uniform Resource Identifiers), which are globally unique.

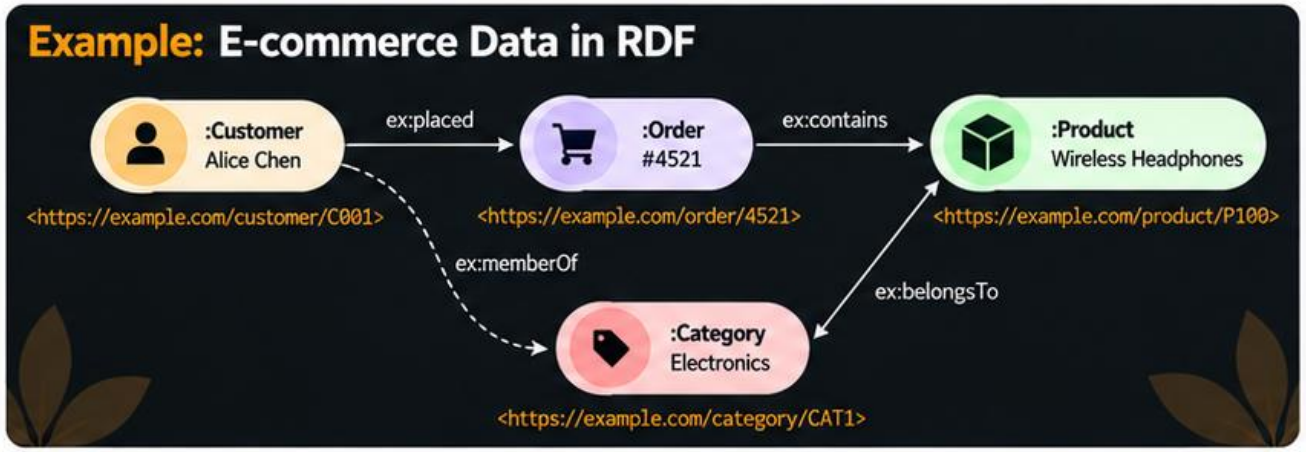
3. **No Rows or Nodes**
   RDF does not have tables, rows or nodes as first-class structures.
   All data is decomposed into atomic triples.

4. **Ontologies Define Meaning**
   Vocabularies (RDFS/OWL) define the terms, their meaning, subclass relationships and inference rules.

5. **SPARQL for Querying**
   SPARQL is the query language for RDF, designed to query graphs of triples.

6. **RDF-star (RDF 1.2)**
   Extends RDF to allow triples themselves to be the subject or object of other triples, enabling "relationship properties".

Example: E-commerce Data in RDF

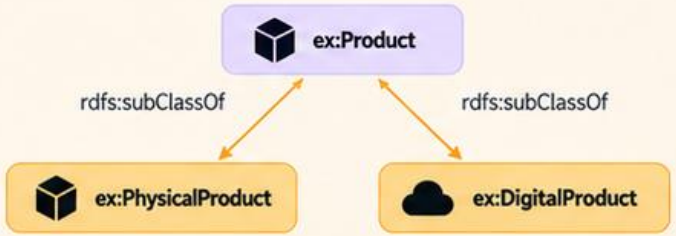


RDF Triples (Statements)

Subject (URI)	Predicate (URI)	Object (URI or Literal)
ex:customer/C001	ex:name	"Alice Chen"
ex:customer/C001	ex:memberOf	ex:category/CAT1
ex:customer/C001	ex:placed	ex:order/4521
ex:order/4521	ex:orderDate	"2024-03-10"^^xsd:date
ex:order/4521	ex:contains	ex:product/P100
ex:product/P100	ex:name	"Wireless Headphones"
ex:product/P100	ex:price	"199.00"^^xsd:decimal
ex:product/P100	ex:belongsTo	ex:category/CAT1
ex:category/CAT1	ex:name	"Electronics"

Ontology (Vocabulary) Example

Ontologies define the meaning of terms and how they relate.



Example axioms (OWL/RDFS)

```
ex:PhysicalProduct rdfs:subClassOf ex:Product
ex:DigitalProduct rdfs:subClassOf ex:Product
ex:price rdf:type owl:DatatypeProperty
ex:belongsTo rdf:type owl:ObjectProperty
ex:Product rdfs:label "Product"
ex:Product rdfs:comment "A product offered for sale."
```

SPARQL Query Example

Find all products purchased by Alice Chen:

```
PREFIX ex: <https://example.com/>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?product ?name ?price
WHERE {
  ex:customer/C001 ex:placed ?order .
  ?order ex:contains ?product .
  ?product ex:name ?name .
  ?product ex:price ?price .
}
ORDER BY ?name
```

Strengths

- Universal interoperability through URI-based identification
- Supports logical reasoning and inference (OWL reasoners)
- Open-world assumption enables integration of partial knowledge
- Ideal for knowledge graphs with heterogeneous data sources
- Linked Data ecosystem (DBpedia, Wikidata, schema.org)
- SPARQL is expressive for federated queries across endpoints

Weaknesses

- Verbose: even simple data requires many triples
- Tooling is less mature than relational or property graph
- SPARQL has a steep learning curve
- RDF triplestores have smaller ecosystems than Neo4j
- Open-world assumption can be surprising for engineers used to closed-world systems
- Performance on certain traversal patterns is less optimized than Neo4j

RDF-star Example (Relationship Properties)

In RDF-star, a triple can be the subject or object of another triple, allowing properties about relationships.

```
<< ex:customer/C001 ex:placed ex:order/4521 >>
   ex:channel "mobile_app" .

<< ex:customer/C001 ex:placed ex:order/4521 >>
   ex:timestamp "2024-03-10T14:22:00Z"^^xsd:dateTime .
```

Key Takeaway

RDF represents knowledge as a set of interconnected triples, using global identifiers and shared vocabularies.
It is designed for data integration, reasoning and the Semantic Web, offering unmatched interoperability and expressiveness.

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The RDF Model

A universal, standards-based model for the Semantic Web — everything is a triple.

TURNING DISCONNECTED DATA INTO MEANINGFUL CONNECTIONS

Core Concepts

1. **Everything is a Triple**
   Data is represented as subject → predicate → object (a statement or fact).

2. **URIs Identify Everything**
   Subjects, predicates, and (resource) objects are identified by globally unique URIs (Uniform Resource Identifiers).

3. **No Rows or Nodes**
   RDF decomposes all data into atomic statements. There are no tables, rows, or first-class nodes.

4. **Ontologies Define the Vocabulary**
   Ontologies (OWL or RDFS) specify classes, properties, subclass relationships, and inference rules (what terms mean).

5. **SPARQL is the Query Language**
   SPARQL is used to query RDF data across local or remote endpoints (federated queries).

6. **RDF-star (RDF 1.2)**
   Allows triples themselves to be the subject or object of other triples, enabling "relationship properties" in RDF.

What is an RDF Triple?

A triple is a simple statement: **Subject** → **Predicate** → **Object**



Multiple Triples for the Same Information

A simple e-commerce example represented as RDF triples:

<ex:C001>	<ex:name>	"Alice Chen"	(literal value)
<ex:C001>	<ex:registeredAt>	"2023-01-15"^^<xsd:date>	(literal value)
<ex:C001>	<ex:purchased>	<ex:O4521>	(resource URI)
<ex:O4521>	<ex:contains>	<ex:P100>	(resource URI)
<ex:P100>	<ex:name>	"Wireless Headphones"	(literal value)
<ex:P100>	<ex:price>	"199.00"^^<xsd:decimal>	(literal value)
<ex:P100>	<ex:category>	<ex:CAT1>	(resource URI)

RDF-star: Properties on Relationships

Standard RDF cannot directly attach properties to a relationship. RDF-star allows triples to be the subject or object of other triples.



SPARQL Query Example

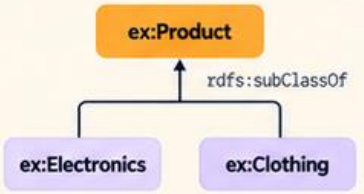
Find all products purchased by Alice Chen:

```
PREFIX ex: <https://example.org/ontology/>

SELECT ?product ?name ?price
WHERE {
  ex:C001 ex:purchased ?order .
  ?order ex:contains ?product .
  ?product ex:name ?name .
  ?product ex:price ?price .
}
```

Ontologies (OWL/RDFS)

Ontologies define the vocabulary and meaning of terms.



Key ontology concepts:

- Classes (e.g., ex:Product)
- Properties (e.g., ex:purchased)
- Subclass relationships (e.g., ex:Electronics ⊑ ex:Product)
- Domain and range constraints
- Inference rules (OWL)

Key Terms (Quick Definitions)

RDF	Resource Description Framework; a W3C standard for representing data as triples.
Triple	A statement: subject → predicate → object.
URI	A globally unique identifier (e.g., https://example.org/C001).
Literal	A concrete value (string, number, date).
Ontology	A formal vocabulary defining classes, properties, and rules (OWL/RDFS).
SPARQL	The query language for RDF.
RDF-star	Extension allowing triples to be subject or object of other triples.

Strengths

- Universal interoperability through URI-based identification
- Supports logical reasoning and inference
- Open-world assumption (partial knowledge)
- Natural for knowledge graphs from heterogeneous sources
- Linked Data ecosystem (DBpedia, Wikidata)
- SPARQL supports federated queries

Weaknesses

- Verbose: many triples even for simple data
- Tooling less mature than relational or property graph
- SPARQL has a steep learning curve
- Smaller ecosystem than Neo4j
- Open-world assumption can be surprising for engineers used to closed-world systems
- Certain traversal patterns are less optimized than property graphs

Key Takeaway

RDF provides a universal, standards-based model for representing and linking **data** on **the web**. It excels at interoperability, reasoning, and integrating diverse data sources, making it the foundation of the Semantic Web and many knowledge graphs.

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When Does Each Data Model Excel?

Choosing the right model for the right problem

1 Relational Model

Best for structured tabular data, transactions, reporting, and aggregate queries

Customers

id	name	city
1	Alice	NY
2	Bob	SF
3	Carol	LA

Orders

id	customer_id	total
101	1	250
102	2	125
103	1	320

Products

id	name	price
10	Laptop	1200
11	Phone	800
12	Headphones	200

Typical question pattern: "What happened in rows, tables, and totals?"

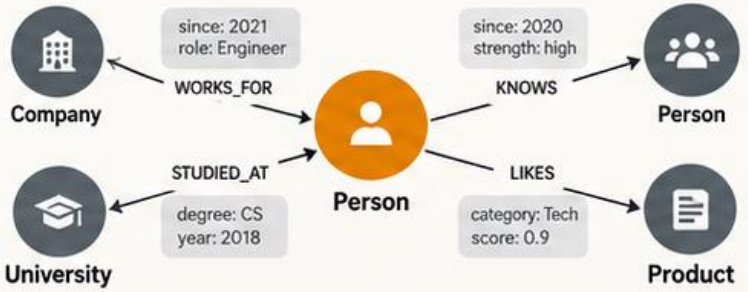
Excels in: Banking systems, ERP, e-commerce orders, inventory, payroll, booking systems, OLTP + many OLAP/ reporting workflows

Why it excels: Strong schema, ACID transactions, mature ecosystem, excellent SQL support

Watch out: Deep relationship traversal and graph-shaped questions become harder

2 Property Graph Model

Best for connected data, traversal-heavy workloads, multi-hop questions



Typical question pattern: "How is A connected to B through many steps?"

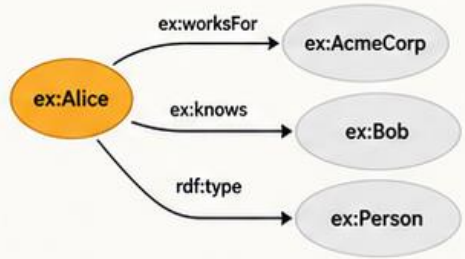
Excels in: Recommendation engines, fraud detection, supply chain dependency analysis, social networks, knowledge navigation, IT dependency graphs

Why it excels: Relationships are first-class, direct traversal, intuitive graph patterns

Watch out: Pure analytics over the whole dataset may not be the best fit

3 RDF Model

Best for semantic interoperability, knowledge integration, shared vocabularies, reasoning




```
@prefix ex:   <http://example.org/>
@prefix foaf: <http://xmlns.com/foaf/0.1/>
@prefix rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

Typical question pattern: "How do we integrate and understand heterogeneous knowledge from many sources?"

Excels in: Enterprise knowledge graphs, linked data, ontology-driven integration, scientific knowledge graphs, data sharing across organizations, Wikidata-like settings

Why it excels: URIs, ontologies, inference, interoperability, semantic web standards

Watch out: More verbose, steeper learning curve, not always ideal for app-style graph traversal performance

How to choose?

A simple rule of thumb

1. **Choose relational when** the world looks like tables and transactions.
2. **Choose property graph when** relationships and paths are the core problem.
3. **Choose RDF when** meaning, interoperability, and shared semantics matter most.

There is no universal winner — the best model depends on the shape of the data and the questions you need to answer.

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Where the Relational Model Excels

Best for structured data, transactions, and large-scale tabular analysis

Choose the relational model when your world looks like tables, records, and clearly defined business transactions.

1 Best-fit use cases

Banking & payments
Account balances, transfers, transaction history.

E-commerce operations
Customers, orders, payments, inventory.

ERP / payroll / HR systems
Employees, salaries, attendance, benefits.

Reservation & booking systems
Airline, hotel, ticketing, appointments.

BI reporting and many OLAP workloads
Sales summaries, monthly totals, dashboards.

2 Relational schema example

A simple e-commerce schema with tables and foreign keys showing how data is related.

Customers

id	name	email	city
1	Alice	alice@ex.com	NY
2	Bob	bob@ex.com	SF
3	Carol	carol@ex.com	LA

customers.id (FK)

Orders

id	customer_id	order_date	total
101	1	2024-03-01	250
102	1	2024-03-04	125
103	2	2024-03-05	320

orders.id (FK)

Order_Items

id	order_id	product_id	quantity	price
1001	101	10	2	600
1002	101	12	1	200
1003	102	11	3	150

products.id (FK)

Products

id	name	category	price
10	Laptop	Electronics	1200
11	Phone	Electronics	800
12	Headphones	Accessories	200

3 Why it excels here

Strong schema enforcement
Clearly defined structure, data types, constraints, and relationships keep data consistent.

ACID transactions
Reliability guarantees for transactions.

ACID = reliability guarantees for transactions.
Example: moving money from Account A to Account B should either fully succeed or fully fail.

Excellent SQL support
Powerful and expressive SQL for querying, reporting, and analytics.

Mature ecosystem and tooling
Wide range of databases, tools, ORMs, BI tools, and operational support.

Very good for aggregations and tabular analysis
Efficient for summarizing large amounts of structured data.

Aggregate query = summarizes many rows.
Example: total sales by month.

4 Typical questions it answers well

- What were total sales by region this month?
- Which customers placed orders this week?
- How many items are left in stock?
- Which employees were paid in March?

Watch out

Deep multi-hop relationship questions and variable-depth traversals become harder and more complex in SQL.
As the number of joins grows, queries can become complex, harder to write, and harder to maintain.

Relational databases excel when correctness, structure, transactions, and tabular analysis are the core needs.

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Where the Property Graph Model Excels

Best for connected data, traversal-heavy questions, and multi-hop reasoning

Choose a property graph when relationships and paths are the heart of the problem.

1 Best-fit use cases

Recommendation engines
People who bought X also bought Y.

Fraud detection
Accounts connected through devices, IPs, or payment methods.

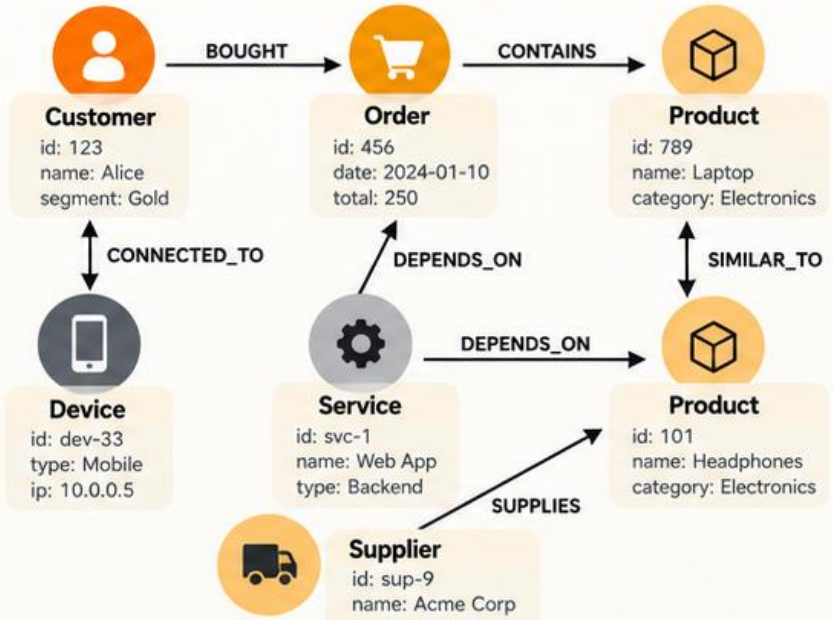
Supply chain dependency analysis
Upstream supplier risk across many tiers.

Social and professional networks
Who knows whom, community detection, shortest paths.

IT operations / dependency mapping
Which services are affected if one server fails.

2 Property graph example

A single, connected view of people, products, orders, services and more, with rich relationships.



3 Why it excels here

Relationships are first-class
Entities and their relationships (with properties) are core to the model.

Traversal is direct and natural
You can follow relationships step by step, just like the real world.

Multi-hop queries are easier to express
You can query across multiple hops in a single, natural statement.

Variable-depth paths are a natural feature
You do not know in advance how many steps you need, and that's OK.

Domain models map well from whiteboard to graph
It's intuitive to model real-world entities and relationships as a graph.

4 Typical questions it answers well

- Which products are connected to what Alice bought through similar customers?
- Which accounts are linked within 3 hops of a suspicious account?
- What is the shortest path between two users?
- Which services will be impacted if this database goes down?

5 Key terms (beginner-friendly)

Traversal	Following relationships step by step.	Example: Alice → bought → product → similar product → other customers.
Multi-hop	More than one relationship away.	Example: supplier → supplier → supplier.
Variable-depth query	You do not know in advance how many steps you need.	Example: find all upstream suppliers at any tier.

Watch out

Pure whole-dataset aggregations and traditional BI-style reporting are often not the strongest use case for property graphs.

Property graphs excel when the meaning of the data lives in the connections, not just in individual records.

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Where the RDF Model Excels

Best for semantic interoperability, knowledge integration, and shared meaning

Choose RDF when many systems, vocabularies, and data sources must work together with shared meaning.

1 Best-fit use cases

Enterprise knowledge integration
Combine data across departments and systems.

Linked Data / web knowledge graphs
DBpedia, Wikidata, schema.org-like ecosystems.

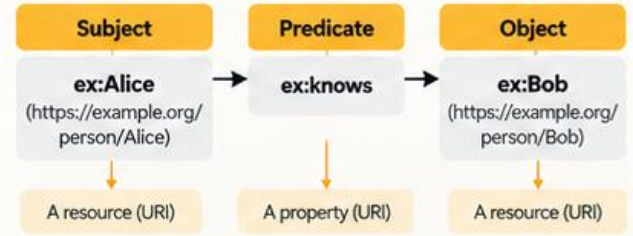
Scientific and biomedical knowledge graphs
Genes, proteins, diseases, publications.

Ontology-driven data integration
Common vocabulary across heterogeneous sources.

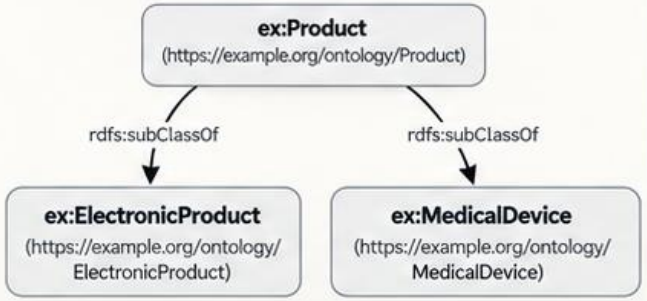
Cross-organization data sharing
Partners exchange data using shared semantics.

2 RDF model example

RDF Triple (Subject → Predicate → Object)



Ontology / Vocabulary Example (Subclass Relationships)



Key Terms

URI
A globally unique identifier. Example:
`https://example.org/product/P100`

Ontology
A formal vocabulary of terms and meanings.
Example: saying that every Doctor is also a Person.

Inference
Deriving new facts from known facts.
Example: if Alice is a Professor and every Professor is a Person, then infer Alice is a Person.

Federated query
One query that can reach multiple remote data sources.

3 Why it excels here

Global identifiers with URIs
Unambiguously identify things across systems, organizations, and the web.

Shared vocabularies and ontologies
Use common, explicit meanings for data, enabling consistent understanding.

Supports reasoning and inference
Derive new facts from existing knowledge using ontologies and rules.

Natural for integrating heterogeneous sources
Designed to connect data from different systems, domains, and formats.

Supports linked data and federated queries
Easily link to external knowledge bases and query across multiple remote data sources.

4 Typical questions it answers well

- How do we **merge** product, supplier, and research data from different systems?
- What **facts** can we infer from the ontology?
- How do we link our data to external knowledge bases?
- Which remote SPARQL endpoints contain related information?

Watch out

RDF can be more verbose, SPARQL has a learning curve, and app-style traversal performance may not always match property graph systems.

RDF excels when interoperability, shared semantics, and knowledge integration matter more than simple app-centric data modeling.